

Product Carbon Footprint Study Report

Product Name: blower and heat exchange unit, Twl-700

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: blower and heat exchange unit, Twl-700

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1Item blower and heat exchange unit, Twl-700

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1Item blower and heat exchange unit, Twl-700.

2. System Boundary

blower and heat exchange unit production, Twl-700, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of blower and heat exchange unit, Twl-700 produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **767.368 kg CO2-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

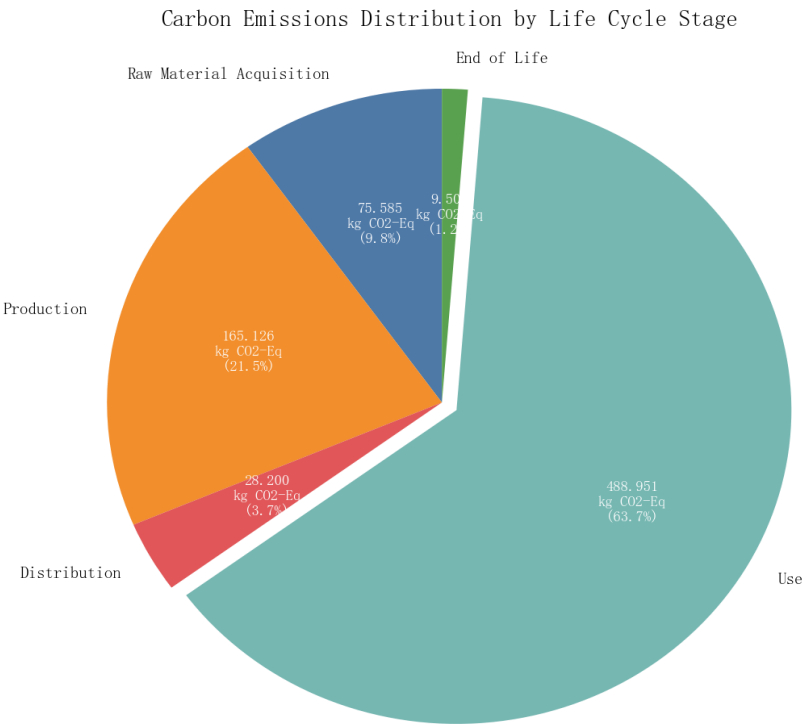


Table 1: Carbon Emissions by Life Cycle Stage

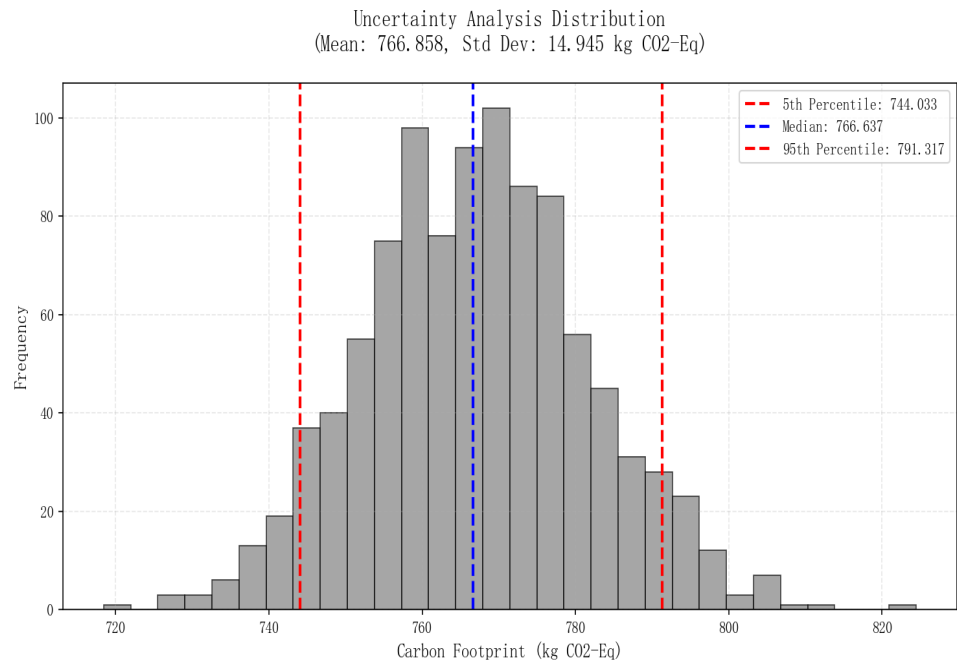
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	75.585	9.85
Production	165.126	21.52
Distribution	28.200	3.67

Use	488.951	63.72
End of Life	9.506	1.24
Total	767.368	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	766.858
Standard Deviation	14.945
Coefficient of Variation (CV)	1.95%
Median	766.637
5th Percentile	744.033
95th Percentile	791.317

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the blower and heat exchange unit, Twl-700, is 767.368 kg CO₂-Eq, a moderate level for such electromechanical equipment, reflecting a typical balance between manufacturing and operational impacts. The lifecycle stage analysis reveals a dominant contribution from the use phase, accounting for 63.7% (488.951 kg CO₂-Eq), indicating that energy consumption during operation is the primary driver of emissions. Production contributes 21.5% (165.126 kg CO₂-Eq), while raw material extraction and processing add 9.8% (75.585 kg CO₂-Eq). Distribution and end-of-life stages are minor, at 3.7% and 1.2%, respectively.

The major emission sources are concentrated in the use phase, likely due to electricity consumption over the product's operational lifetime. Production emissions also represent a significant share, stemming from manufacturing processes and energy use. The low contributions from distribution and end-of-life suggest that transport and disposal have limited environmental impact. However, the top process contributions are reported as "Unknown" with zero values, indicating a lack of process-level granularity in the current assessment, which limits the identification of specific emission hotspots within each stage.

2. Statement of Assumptions and Limitations

This LCA study is based on several key assumptions. The use phase assumes a standard operational profile, including average load conditions and a typical product lifespan, with electricity sourced from a regional grid mix. The production stage assumes average manufacturing efficiency and generic material inputs, while raw material data rely on industry-average databases. End-of-life assumes conventional disposal methods, such as landfilling or recycling, without specific recovery rates.

Limitations include the absence of process-level data for the top contributing processes, as all are marked "Unknown," likely due to data gaps or reliance on aggregated datasets. This reduces the ability to pinpoint exact emission sources and may underestimate certain impacts. Data quality is further constrained by the use of generic secondary data, which may not fully reflect the specific supply chain or production technologies of the manufacturer. The system boundary is cradle-to-grave, covering raw material extraction, production, distribution, use, and end-of-life, but excludes capital goods, infrastructure, and potential maintenance impacts. The uncertainty analysis (CV 1.95%, 90% CI 744–791 kg CO₂-Eq) indicates relatively low variability, lending confidence to the overall result, but does not compensate for missing process details.

3. Suggestions for Improvement

- ****Improve use-phase energy efficiency****: Implement high-efficiency motors, optimized heat exchange designs, and variable speed drives to reduce electricity consumption during operation. This directly targets the largest emission stage (63.7%) and offers the highest reduction potential.
- ****Optimize production processes****: Transition to renewable energy sources (e.g., solar or wind) for manufacturing facilities, and adopt lean production techniques to minimize material waste and energy use. This can cut the 21.5% share from production.

- ****Enhance raw material sustainability****: Substitute virgin metals and plastics with recycled or bio-based materials where feasible, and work with suppliers to reduce upstream emissions. This addresses the 9.8% from raw materials.
- ****Improve data quality for process contributions****: Conduct a detailed process-level inventory to replace the "Unknown" entries. Engage suppliers to provide primary data on key components (e.g., motors, fans, heat exchangers) to enable targeted improvements and reduce uncertainty.